



Augmenting Common Spatial Patterns to deep learning networks for improved alcoholism detection using EEG signals

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ABSTRACT

One of the main risk factors for numerous health problems is excessive drinking. Alcoholism is a severe disorder that can affect a person's thinking and cognitive abilities. Early detection of alcoholism can help the subject regain control over their drinking habits and help them recover faster. The diagnosis of alcoholism is challenging, primarily when standard diagnostic tests rely on blood tests and questionnaires that are subjective to the patient and the examiner. This poses the need for fast, reliable, automatic, and preferably non-invasive ways to detect alcoholism. A non-invasive method to capture the electrical activity of the brain is the electroencephalogram (EEG). It can help detect alcohol use disorders in a subject. To our knowledge, no previous work in the literature proposes a mechanism to detect alcoholism using an EEG signal using a deep learning model that considers the spatio-temporal nature of EEG signals. Two works in the literature using a deep learning architecture are Fayyaz et al. (2019) and Farsi et al. (2021). However, both use a simple deep learning architecture and ignore the spatio-temporal nature of multichannel EEG signals. This paper suggests a new hybrid architecture called CSP-CNN-LSTM-ATTN. To obtain nonlinear and nonstationary spatio-temporal features from EEG signals that can accurately classify them as either alcoholic or control (not alcoholic). Our method integrates Common spatial patterns (CSP) for feature extraction, convolutional neural networks (CNN) for spatial representation, long-short-term memory (LSTM) for temporal learning, and attention networks (ATTN) for feature weighting, enhancing classification performance. Experiments on the publicly available UCI EEG dataset demonstrate that our model achieves state-of-the-art results, outperforming existing methods with an accuracy of 98.60%, F1-score 98.59, recall 98.24, precision 98.95, MCC 97.20, and AUC 99.64. The source code for this study is available at Github repository <https://github.com/n28neeraj/Alcoholism-Detectio>.

1. Introduction

Alcoholism is a psychological disorder in which the patient loses the ability to control alcohol intake. This disorder is also termed Alcohol Use Disorder (AUD). This disorder also severely affects the brain and cognitive abilities of the subject. Alcoholism can also lead to extreme emotional and behavioral changes that affect the inner circle and the social circle of the alcoholic person. Chronic alcohol consumption can cause memory loss and can also inhibit the patient's decision-making ability. Excessive alcohol consumption can result in a variety of permanent health disorders. Alcohol consumption adversely affects the communication pathways of the subject's nervous system and negatively hampers their brain function. Some notable short-term alcohol consumption symptoms are slurred and unclear speech, mood swings, impairment of vision, and improper neural coordination. Apart

from the brain and nervous system, alcohol consumption can also inhibit the functioning of other body organs like the liver, pancreas, heart, and immune system, leading to diseases like cardiac arrests, high blood pressure, fibrosis, and cancer.

The detection of alcoholism requires fast and reliable methods. If it goes undetected, it can adversely affect the patient's life and indirectly affect his friends and family. There is a lack of cost-effective, noninvasive, reliable, and fast methods to detect alcoholism. To our knowledge, analyzing the symptoms reported by the subject is the primary method for detecting alcoholism. The questionnaire-based approach is another strategy similar to that used by medical professionals. These approaches require manual intervention and are highly subjective regarding how well and honestly the subject describes their condition. Furthermore,

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physical symptoms can differ from subject to subject, leading to variations and errors. These factors make these approaches error-prone and unreliable. Manual analysis of each topic is time-consuming. Another approach is to use the blood test for the detection of alcoholism. Although more reliable than the earlier questionnaire-based methods, this method is invasive, costly, and time-consuming.

The method to measure brain electrical activity is called an electroencephalogram (EEG). EEG measures changes in electrical voltage using metal disk electrodes positioned on the subject's scalp at various locations. Our brain's neurons convey messages to many parts of our bodies through electric signals. These electrical signals can be weakly sensed on our scalp. EEG is a well-known technique frequently used to identify many diseases and abnormalities in the brain. Scientists and medical professionals have used EEG signals to identify alcoholism. EEG analysis with deep learning methods for detecting normal and abnormal brain states has been extensively investigated recently [1–3]. These experts argue that the electrical activity of the brain in healthy and alcoholic patients is different. Medical professionals now distinguish between alcoholic and non-alcoholic people using established patterns in EEG waves. Manual analysis of EEG signals is complex. High background knowledge and practice are required to recognize the delicate patterns in these EEG signals and make a correct judgment. Moreover, the EEG data used for medical purposes is not single-channel and is highly prone to noise. This EEG data is multidimensional as it is collected from multiple channels. The multidimensional and noisy nature of EEG time series data makes interpreting these signals harder and more error-prone. Even highly skilled medical practitioners can miss delicate patterns in the data due to noise. Manual analysis of such a vast amount of noisy data would be time-consuming and costly. Hence, there is a severe need for systems that can automate the task of alcoholism detection.

Much research on automatic alcoholism detection suggests that statistical methods and machine learning models can be used to use EEG signals to find people who are drinking too much. However, most of these techniques propose handcrafted features that require manual intervention and domain-level knowledge. Acharya et al. [4] acknowledged that EEG signals are nonstationary, complex, and nonlinear. Hence, statistical and frequency domain methods are used sparingly in classifying EEG signals. After getting different nonlinear features, they used a Support Vector Machine (SVM) classifier with different kernel functions to classify the data. They used Sample Entropy (SampEn), Approximate Entropy (ApEn), Largest Lyapunov Exponent (LLE), and some other higher-order spectra features for their classification task. They achieved an accuracy of 91.7%. However, they used only 60 samples—30 alcoholic and 30 non-alcoholic samples. Kumar et al. used only 40 samples (20 alcoholics and 20 controls) and reported an accuracy of 88%. They used peak power frequency, relative power frequency, and absolute power frequency for feature extraction, and for classification, they used SVM. Shri et al. [5] studied 20 samples, 10 of whom were alcoholics and 10 of whom were not. The Analysis of Variance (ANOVA) test was used to rank the entropy parameters (ApEn), and support vector machines (SVM) and back propagation neural networks (BPNN) were set up. They found that it worked 90% of the time. Kannathal et al. used 60 data samples (30 from alcoholics and 30 from controls) and got 90% of the time right when they said whether the EEG signals came from an alcoholic or a control subject. They extracted parameters, including the correlation dimension, entropy, maximum Lyapunov exponent, and Hurst exponent. Faust et al. [6] used high-order spectra cumulants to extract features from an EEG time series. They claimed that the Fuzzy Sugeno Classifier (FSC) had the highest accuracy of 92.40% and the best performance on the extracted HOS features. Zhu et al. [7] applied custom features such as SampEn and the horizontal visibility graph entropy (HVGE). They also used statistical analysis methods to choose the optimal channels and then used KNN and SVM for classification. It shows that choosing appropriate channels and extracting the correct features can help achieve

a high accuracy of about 95.8% using 1200 samples. However, it used handcrafted features and statistical analysis, which might not be generalized to other cases. One interesting observation about some of this literature is that they used statistically handcrafted features and machine learning algorithms for classification. Using handcrafted features is a significant drawback, as it requires domain-level knowledge. Many of these works use a limited number of samples; therefore, the results of these works may only generalize well in some cases. Moreover, there needs to be more deep learning techniques in this domain. Many works discussed above claim that multichannel EEG signals are noise-prone, nonstationary, and nonlinear. Deep learning models have worked well with nonlinear and noise-prone data in several domains. To diagnose alcoholism using EEG signals, Fayyaz et al. [8] and Farsi et al. [9] have attempted to apply deep learning; however, both of these works only consider the temporal component of EEG signals. Fayyaz et al. used peak visualization to extract features, and Long Short Term Memory (LSTM) was used to make the final classification. To categorize EEG signals as alcoholic or control, Farsi et al. [9] used LSTM and a combination of principal component analysis (PCA) and artificial neural networks (ANN). However, both attempts have used only temporal features of the EEG signal and have entirely ignored the spatial aspect of the multichannel EEG signals; hence, their limited success.

Multichannel EEG signals are spatio-temporal, i.e., they contain spatial and temporal features. Moreover, these multiple channels can be challenging and complex due to the correlation between time series data sampled by different nearby channels. In the literature, such problems are dealt with through mechanisms like PCA and Common Spatial Patterns (CSP). CSP is widely used in tasks of the brain-computer interface (BCI) using EEG signals, as in [10,11]. According to these works, preprocessing of the data using CSP improves the discriminability of the classes. Reduces high-dimensional data to the lower-dimensional subspace, making the data easier to classify.

To our knowledge, no previous work in the literature proposes a mechanism to detect alcoholism using an EEG signal using a deep learning model that considers the spatiotemporal nature of EEG signals. A few works using a deep learning architecture in the literature are [8,9,12].

Using EEG as a non-invasive method of examining cerebral activity, deep learning networks have been widely used in the diagnosis of brain diseases such as alcoholism and Alzheimer's disease. Research on acupuncture-induced brain modulation [13,14] has shown that EEG-based investigations help evaluate neuromodulation effects and functional connectivity. Previous studies have shown encouraging results in the diagnosis of Alzheimer's disease, such as supervised fuzzy learning of EEG signals [1]. Building on these developments, our research estimate functional brain networks with the goal of improving the precision and interpretability of deep learning-based EEG-based diagnosis.

However, both use a simple deep learning architecture and ignore the spatio-temporal nature of multichannel EEG signals. In addition, advanced statistical techniques such as CSP, which have proven effective in similar classification tasks, have never been used in the detection of alcoholism using EEG signals. This study proposes a model that uses CSP and a combination of convolutional neural networks (CNN), LSTM, and Attention networks to extract spatiotemporal features from multichannel EEG signals and accurately classify them as alcoholic or non-alcoholic. This novel architecture CSP-CNN-LSTM-ATTN network takes an entire EEG signal matrix as input, extracts latent spatiotemporal characteristics from the input EEG matrix, and then categorizes the signal as alcoholic or control. The proposed strategy performs better than various methods already published in the literature and provides excellent accuracy.

The paper is structured as follows: The data set utilized for the work is described and analyzed in Section 2; the approach of the proposed architecture is explained in Section 3. The experimental setup, results,

Table 1

Dataset statistics.

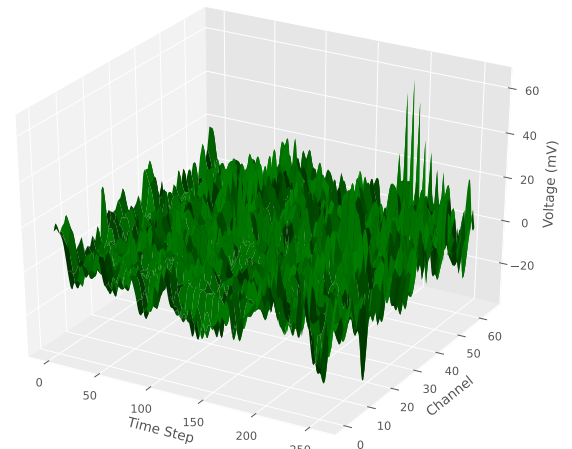
Statistic	Alcoholic	Control
Number of trials	600	600
Mean	-1.3870	-2.0293
Median	-0.956	-1.5870
Standard deviation	8.9820	9.9376
Minimum	-187.907	-213.4909
Maximum	293.091	453.847
Skewness	0.5319	1.4117
Kurtosis	30.0946	42.8731
Quartile 1	-5.2800	-7.0190
Quartile 3	2.7770	3.0820
Inter quartile range (IOR)	8.0570	10.1010

and performance evaluation of the proposed architecture to the chosen data set comparing previous research in the field are covered in more detail in Section 4. Section 5 presents some concluding remarks on this work.

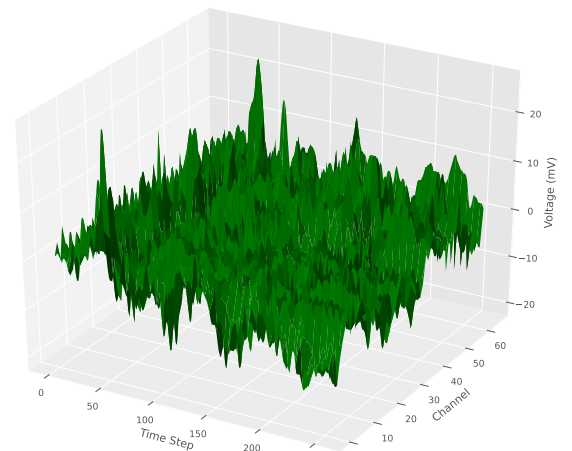
2. Data description

We used a publicly available dataset for this investigation that offers EEG correlates of genetic propensity to alcoholism. The data set was downloaded from the Center for Machine Learning and Intelligent Systems repository at the University of California, Irvine <https://archive.ics.uci.edu/dataset/121/eeg+database> [15]. The studies covered in Section 1 extensively used this dataset. These data were provided by the Neurodynamics Laboratory at the State University of New York Health Center in Brooklyn. Samples from 122 people are included in the original dataset. Multichannel EEG recordings from alcoholic and control patients are included in this collection. Three different stimuli were used in 120 trials for each person. Sixty-four electrodes were placed in various locations on the scalps of the patients to capture the data. The American Electroencephalographic Association (1990) standard electrode location nomenclature specifies a number of standard places for placing electrodes on the head. Each of these electrodes was used to sample the electrical voltage at the scalp to capture the brain's electrical activity at 64 standard sites with a frequency of 256 Hz. Going by this, each electrode sampled voltage readings every 3.9 ms. Each trial was recorded for 1 s. Therefore, each trial yielded data in a two-dimensional matrix of dimensions 64×256 , where 64 denotes the number of channels or electrodes, and 256 is the number of time samples collected in one second. A three-dimensional view of these data is provided in Fig. 1. The vertical axis reports the voltage value in millivolts (mV), one of the horizontal axes represents time, and the horizontal axis represents channel number. For a clearer view of EEG signals, we plot the first eight channels of an alcoholic sample and a control sample on a two-dimensional graph as shown in Fig. 2.

The provided dataset is available in three formats from the UCI dataset: small, large, and full. In this work, a large dataset is used for analytical purposes. The training set and the test set are the two components of this dataset. Each set has 600 samples. We used a total of 1200 samples for this study. There are 300 alcoholic samples and 300 control samples in each of the training and test sets. The training set includes samples from 10 subjects who were alcoholics and 10 controls. 30 samples were added to the training set for each individual. The test set of the vast data also includes 600 samples, including 30 trials from each of 10 alcoholic and 10 control patients. This yields equally sized training and test sets with dimensions of $600 \times 64 \times 256$, where 600 is the total number of data samples in each dataset, 64 is the total number of channels in each sample, and 256 is the total number of time steps for which each sample was recorded. A variety of statistics regarding the sizeable dataset are provided in Table 1.



(a) Control



(b) Alcoholic

Fig. 1. 3D plot of EEG signals.

3. Methodology

As discussed in Section 1, the literature lacks advanced deep learning techniques for detecting alcoholism using EEG signals despite the massive success of deep learning in several related fields. In this section, we aim to propose a method to detect alcoholism in subjects using EEG signal data with the help of advanced deep learning architectures. Although some attempts to use deep learning for this task were made by Fayyaz et al. [8] and Farsi et al. as discussed in Section 1, these pieces of literature used very simple deep learning architectures like LSTM and ANN. The major drawback of these architectures is that they cannot exploit the fact that multichannel EEG signals are Spatiotemporal. These models focused only on temporal features. This section proposes a model that extracts Spatio-temporal features from multichannel EEG signals and classifies these signals as alcoholic or control. The proposed architecture uses the classification task of the CSP, CNN, LSTM, and Attention networks. Below is a brief introduction to the CSP, CNN, LSTM, and Attention networks.

3.1. Common spatial patterns

CSP is a method that linearly transforms a multichannel signal to a lower-dimensional subspace using special matrices called spatial filters. A spatial filter matrix is a matrix in which each row consists of the importance scores assigned to each signal channel. Applying

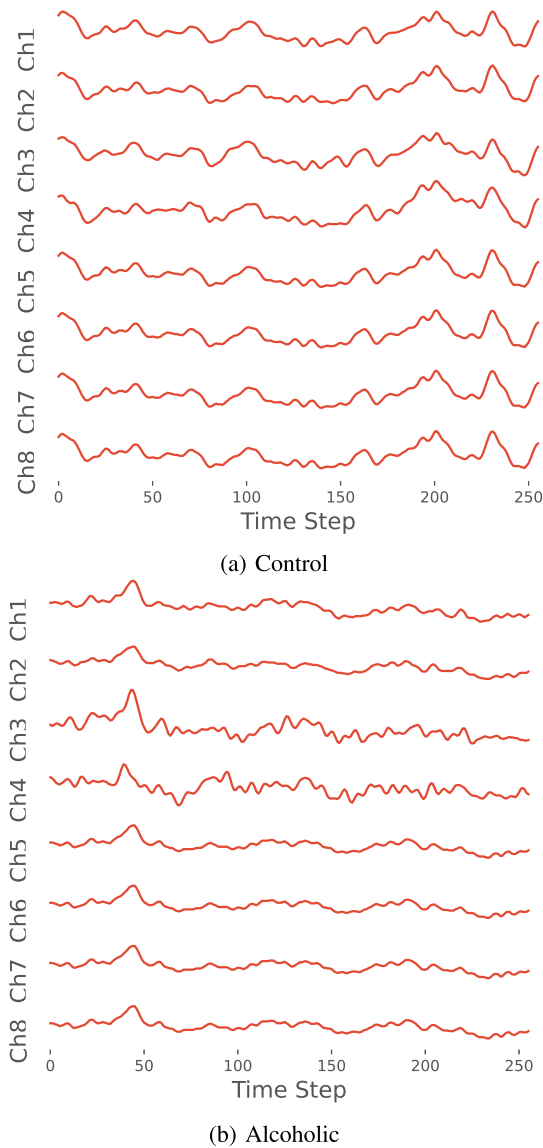


Fig. 2. 2D plot of first 8 EEG signal channels.

CSP aims to maximize the discriminability of two classes and reduce high-dimensional data to lower-dimensional subspace.

Suppose EEG signal matrix A_i is of dimension $H \times T$, where H is the number of channels, and T is the number of time steps for which the sample was recorded. Suppose that these EEG signals are classified into two classes p and q with N_p samples in class p and N_q samples in class q . First, we calculate the transpose of each EEG sample given by A_i^T and the trace of $A_i \cdot A_i^T$. The trace of a matrix is given as the sum of its diagonal elements. Next, we calculate each EEG sample's spatial covariance R_i using (1). For each of the class p and q , we find representative spatial covariance R_p and R_q , respectively, taking the average spatial covariance of all samples belonging to that class. The sum of representative spatial covariance of both classes gives the composite spatial covariance R_c as given by (2)–(4).

$$R_i = \frac{A_i \cdot A_i^T}{\text{trace}(A_i \cdot A_i^T)} \quad (1)$$

$$R_p = \frac{\sum_{A_i \in p} R_i}{N_p} \quad (2)$$

$$R_q = \frac{\sum_{A_i \in q} R_i}{N_q} \quad (3)$$

$$R_c = R_p + R_q \quad (4)$$

The matrix R_c can be decomposed using diagonalization, by calculating its eigenvectors and eigenvalues as shown in (5).

$$R_c = X \cdot \lambda_c \cdot X^T \quad (5)$$

where X is the matrix formed by eigenvectors and λ_c is a diagonal matrix having eigenvalues on the main diagonal in a non-descending order. Using X^T and λ_c we calculate Y which is known as the whitening transformer, by using (6). Y is used to average the spatial covariance matrices for both classes as shown in (7) and (8)

$$Y = \sqrt{\lambda_c^{-1}} \cdot X^T \quad (6)$$

$$Z_p = Y \cdot R_p \cdot Y^T \quad (7)$$

$$Z_q = Y \cdot R_q \cdot Y^T \quad (8)$$

After applying the above linear transformations, sum of eigenvalues of Z_p and Z_q is 1, and both of them yield the same eigenvectors. Next, we find a matrix Q such that it consists of eigenvectors of any one of Z_p or Z_q . To get the final projection matrix or spatial filter W we use (9).

$$W = Q^T \cdot Y \quad (9)$$

3.2. Convolutional neural network

CNNs are widely used in pattern recognition and image classification problems. These networks have shown massive success in recognition and image classification problems due to their remarkable ability to capture and extract localized spatial features from data. The three major components of a CNN architecture are convolution, pooling, and fully connected layers. The convolution layer uses convolution filters and traverses the input data to calculate the scalar product for each local region. The pooling layers downsample the data with spatial dimensions and better capture the spatial features of a localized region. The main aim of CNN architecture is to extract spatial features from data. The spatial features are then converted into feature vectors, which are used as inputs for the fully connected layers to perform the final classification task. The proposed architecture uses CNN for spatial feature extraction as these spatial features can provide vital information to classify the multidimensional EEG signals and should not be ignored.

3.3. Long short term memory

LSTM network is a special type of recurrent neural network (RNN) that overcomes vanishing and exploding gradients faced by traditional RNNs. These networks have cells, each using gates and feedback loops to retain important past information and capture long-range dependency. LSTM uses three types of gates: forget, input, and output. Each gate is used to make different decisions. The input gate decides which values from the input data must be used in computations to calculate the next memory state. Forget gate decides which past information is required or not required anymore. Similarly, the output gate controls the output based on the input and feedback from the last time step. LSTMs have shown a remarkable ability to extract temporal features from time-series data and are widely used for regression and classification. Multi-channel EEG signals are spatial as well as temporal in nature. The temporal features of the EEG signal are of great importance for each channel; the EEG signal is inherently a time series. The proposed work uses LSTM to extract temporal features from the EEG data.

3.4. Attention mechanism

LSTM networks can help capture long-range dependence from time series. However, these networks do not have any mechanism to assign different importance to different parts of the time series. Certain signal parts are more important in signal processing than others, and a specific class is assigned to the signal. For this study, multi-channel noisy EEG signals are classified. The Attention mechanism proposed initially by Bahdanau et al. [16] is needed to deal with this task effectively. The attention layer takes a set of feature vectors say $v_1, v_2, v_3 \dots v_N$ as input. The attention encoder encodes these input vectors into information vectors $i_1, i_2 \dots i_N$. The attention mechanism requires the computation of a set of context vectors. The context vectors are computed using the weighted sum of the encoder output. Computation of these context vectors $c_1, c_2, c_3, \dots c_N$ is done using (10).

$$c_t = \sum_{i=0}^N a_i \cdot i_t \quad (10)$$

Here x_t represent the encoded information vector at time step t . The term a_i denoted the attention scores, calculated using (11) and (12).

$$\text{softmax}(a_i) = \frac{\exp(q_i)}{\sum_{i=0}^N \exp(q_i)} \quad (11)$$

$$q_i = F_{fcn}(i_t, p_{t-1}) \quad (12)$$

The function F_{fcn} in (12) denotes a feed forward network. i_t denotes the encoded information vector and p_{t-1} gives the vector representing the previous cell state. For each time step, the output of attention layer y_t , is calculated using output of the previous time step y_{t-1} , context vector c_t and previous cell state vector p_{t-1} .

3.5. Proposed architecture

Detection of alcoholism requires processing multi-channel EEG data. This data is hard to process because of its high dimensionality and noise. High-dimensional data deteriorates the prediction abilities of the classifier used. Another factor that reduces the prediction accuracy is that data from nearby channels are prone to high correlation. In the proposed architecture, CSP is used to deal with this problem of high dimensionality and correlation. The application of CSP not only linearly transforms a data matrix into a lower-dimensional subspace but also maximizes the discriminability of two classes. Assume that each EEG signal matrix E_i has dimension $C \times T$ where C is the number of channels in the EEG signal, and T is the number of time samples for each channel. The output after applying CSP to each data matrix E_i is processed matrix P_i having dimension $C' \times T$ where C' is the dimension of subspace to which the data is projected, and T is the number of time samples. Each data matrix C_i is broken into smaller segments. C' In this case, the hyper-parameter value should be less than C . We used grid search for experimental purposes to determine the appropriate value of C' . After applying CSP, each processed matrix is divided into smaller segment matrices using the moving-window technique. This process of breaking each processed matrix P_i into smaller segment matrices S_{ij} is done so that each segment matrix covers data for a fixed set of time steps for each channel. CSP and this moving window transformation act as the processing step required to prepare the input for the classifier. Fig. 3 shows the pre-processing pipeline process. The segment matrices S_i generated after pre-processing steps have dimensions $C' \times W$ where W is the moving window size determined using grid search. These segment matrices are passed through the CNN layers distributed over time to extract spatial features. Each segment matrix S_i is sequentially passed through CNN to obtain the feature vectors representing spatial information about the signal. Each segment matrix S_i is converted to a corresponding feature vector v_i using the CNN layer. These feature vectors $v_1, v_2, v_3, \dots v_N$ are then used as input for the LSTM layer, which is used for extraction of temporal features from the data. These LSTM

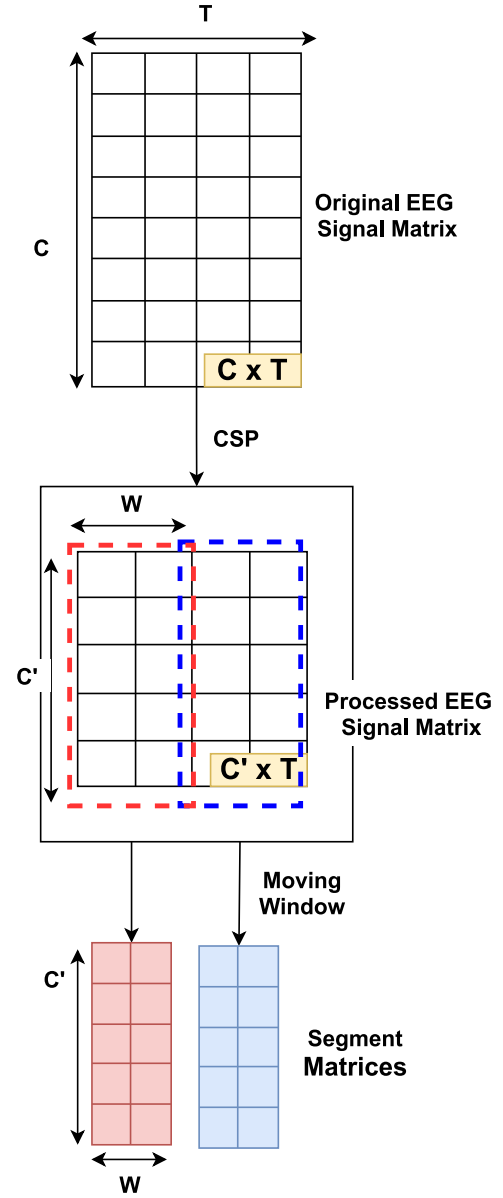


Fig. 3. Pre-processing.

layers transform the features vectors $v_1, v_2, v_3, \dots v_N$ to corresponding feature vectors $s_1, s_2, s_3, \dots s_N$. This set of feature vectors is then fed to the Attention layer for processing. The Attention layer calculates the context vectors and attention scores corresponding to these feature vectors for internal computations as described in Section 3.4. After all the computations, the attention layer finally outputs a single feature vector, passed through the feed-forward neural network to output the final label for the EEG data matrix. The end-to-end architecture for the proposed model is shown in Fig. 4. To summarize, the proposed model first reduces the dimensionality of the EEG data matrix using CSP and then uses the moving window technique to create temporal segment matrices fed as input to a CNN-LSTM-ATTN network. The network extracts Spatio-temporal features from the data to predict whether the input EEG signal is recorded for a control or alcoholic subject. The model's prediction accuracy is higher than most existing approaches present in the literature.

The spatiotemporal features extracted using CNN-LSTM-ATTN architectures for EEG-based alcoholism detection present several advantages over traditional EEG features, such as Power Spectral Density

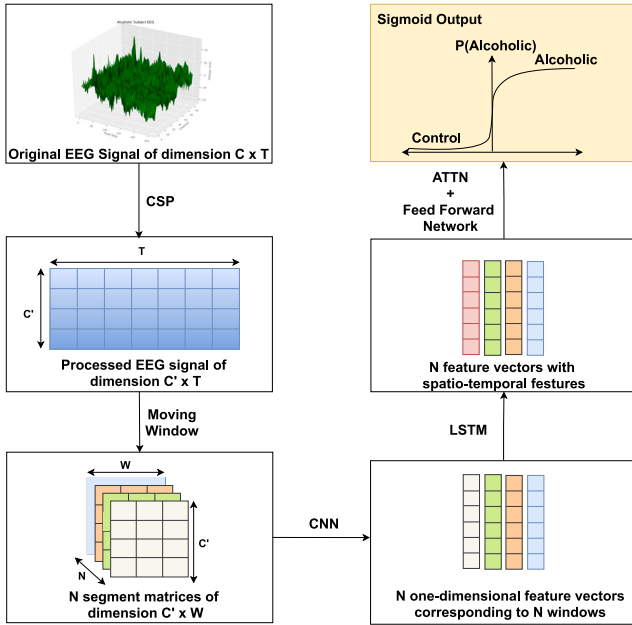


Fig. 4. Overview of the proposed architecture.

(PSD) and functional brain networks. Unlike PSD, which captures spectral power without providing spatial or temporal insights, spatiotemporal features allow a more comprehensive representation of EEG dynamics. By leveraging CNNs, the proposed method captures local spatial dependencies across electrodes, while the LSTM-ATTN network captures temporal dependencies within EEG sequences, providing a richer analysis of how brain signal sequences evolve. This combination enables the detection of subtle spatiotemporal patterns that correlate more directly with conditions like alcoholism. Additionally, while features like PSD and functional connectivity have robust biological interpretability, as shown in studies investigating brain modulation through acupuncture, they do not capture complex temporal dynamics and spatial patterns. The novelty of the proposed method lies in its ability to jointly model spatial and temporal relationships in EEG signals, which is particularly useful in distinguishing nuanced differences in brain activity that are less accessible to static or frequency-domain analyses alone.

4. Results and discussion

This section discusses the experimental studies conducted for the proposed work and the results obtained after applying the proposed model to the data set considered in Section 2. It also shows how different hyperparameters used in the model can affect the proposed model's performance. It describes how the results are affected if different model variations are used along with CSP. Lastly, the results obtained by the proposed model are compared with existing techniques to detect alcoholism using EEG signals.

As described in Section 2, the dimension of the training set is $600 \times 64 \times 256$. CSP is applied to the entire training set.

The training set and test set are projected using the spatial filters obtained from CSP to a lower-dimensional subspace of dimensions $600 \times C' \times 256$; where C' is a hyperparameter that needs to be tuned using grid search. After applying CSP transformation in training and testing, the moving window technique breaks large processed data matrices into smaller segment matrices. The size and stride used for applying moving windows are also hyper-parameters determined using a grid search algorithm. The performance of the proposed model with different C' and window sizes is given in Table 2 in terms of precision.

Table 2
Grid search using accuracy metric.

CSP subspace size (C')	4	8	16	32
Window size				
4	81.83	93.50	95.16	98.00
8	84.83	90.00	95.33	97.83
16	84.16	93.00	96.33	98.00
32	83.50	95.16	96.83	98.66
64	85.83	94.66	95.83	97.83

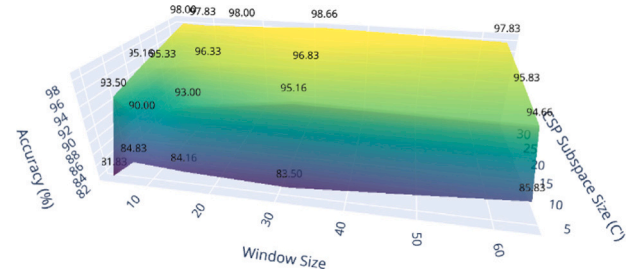
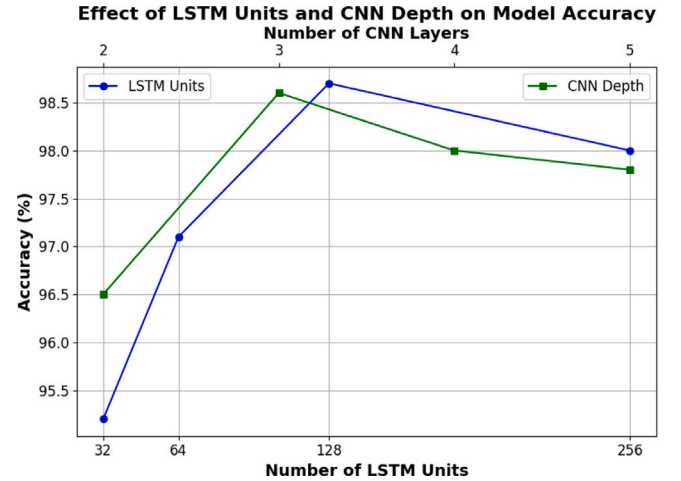
Fig. 5. Accuracy for different window sizes and C' .

Fig. 6. Effect of LSTM units and CNN depth on model accuracy.

Fig. 5 plots the accuracy of proposed architectures for different values of C' and window size.

Table 2 suggests the optimal window size and subspace dimension C' , used in CSP, to be 32. These hyper-parameters values, with moving window using a stride of 2 redefine the dimensions of the training set and test set to be $600 \times 112 \times 32$; where 600 is the number of samples in the training set and the test set, 112 denotes the number of windows in each sample, 32 is the number of channels dimensions after applying CSP and the size of each window.

To comprehensively analyze the model's hyperparameters, we have combined the LSTM units and CNN depth evaluation into a single Fig. 6. The left panel of Fig. 6 shows the effect of varying the number of LSTM units (32, 64, 128, 256) on accuracy, where 128 units yielded the highest performance. The right panel illustrates the impact of different CNN depths (2, 3, 4, 5 layers) on accuracy, with three layers being optimal. This combined figure allows a more streamlined comparison of how these critical hyperparameters affect the overall model performance.

As described in Section 3, each sample is fed into the proposed CNN-LSTM-ATTN network, where the time-distributed CNN layer processes one window at a time, yielding a feature vector for each window. This structure's first set of Conv2D layers has 32 filters with a kernel size of 3×3 and utilizes ReLU activation. This choice of parameters

Table 3

Performance evaluation of the proposed model based on 30 trials.

Metric	Mean	Median	Standard deviation	Min	Max
Accuracy	98.60	98.66	0.24	98.00	99.00
F1-score	98.59	98.65	0.25	97.97	99.00
Recall	98.24	98.33	0.57	96.66	99.33
Precision	98.95	98.99	0.36	98.01	99.32
MCC	97.20	97.33	0.49	96.03	98.00

Table 4

Confusion matrix.

		Actual class	
		Alcoholic	Control
Predicted class	Alcoholic	297	3
	Control	5	295

allows the layer to capture fine-grained spatial features within each 32×32 segment, identifying localized patterns across each window while keeping the computational load manageable. Following this, a Max Pooling 2D layer with a pool size of 3 reduces the spatial dimensions, enabling the network to focus on the most prominent spatial features while reducing complexity. A second set of Conv2D layers, also with 32 filters and a 3×3 kernel, further deepens the feature extraction process and is followed by a Max Pooling 2D layer with a pool size of 2, which refines the feature maps further. After these layers, the grid is flattened into a 1D vector, which feeds into a set of feed-forward layers with 128 and 64 units, respectively, and ReLU activation. This fully connected layer configuration enhances the network's ability to learn complex spatial feature representations for each window, capturing more abstract patterns beyond the raw spatial data. These feature vectors are then input to the LSTM layers, where a set of stacked LSTM layers with 128 units each are configured, allowing the model to capture sequential dependencies across the 112 windows. The choice of LSTM layers, rather than simpler RNNs, facilitates the learning of long-term temporal dependencies essential for sequences of this length and complexity. The output from the LSTM layers is passed to an Attention layer, which outputs a single feature vector that prioritizes more relevant time steps in the sequence, aiding the model's interpretability and improving accuracy by emphasizing critical segments of each sample. This attention-weighted feature vector is then processed through a fully connected Dense layer with 64 units and ReLU activation, refining the temporal-spatial representation before classification. Finally, a Dense layer with a single unit and sigmoid activation generates the binary label for each sample, with binary cross-entropy loss, and the Adam optimizer is used to train the model effectively for binary classification. All the above parameters are optimized using a grid search approach to explore constrained parameter space and identify optimal configurations systematically. Fig. 7 illustrates the change in signal matrix dimensions at each processing step, highlighting the model's capability to capture and refine spatial and temporal features efficiently.

Table 3 shows the statistical results after running the model 30 times.

The results of deep learning models vary on each run due to the stochastic nature of initial weights and other model parameters. To verify the robustness of the model against different initial conditions, we ran the model 30 times and reported the stats obtained from these 30 trials in Table 3. Table 4 gives the confusion matrix obtained after applying the proposed model to the test set. Table 5 summarizes various models used in literature for predicting alcoholism using EEG signals. This table also compares and contrasts these models with the proposed model.

The results obtained by the proposed technique are noteworthy for the existing methodologies adopted for the same dataset. A detailed comparative analysis with the latest methods is provided in Table 6.

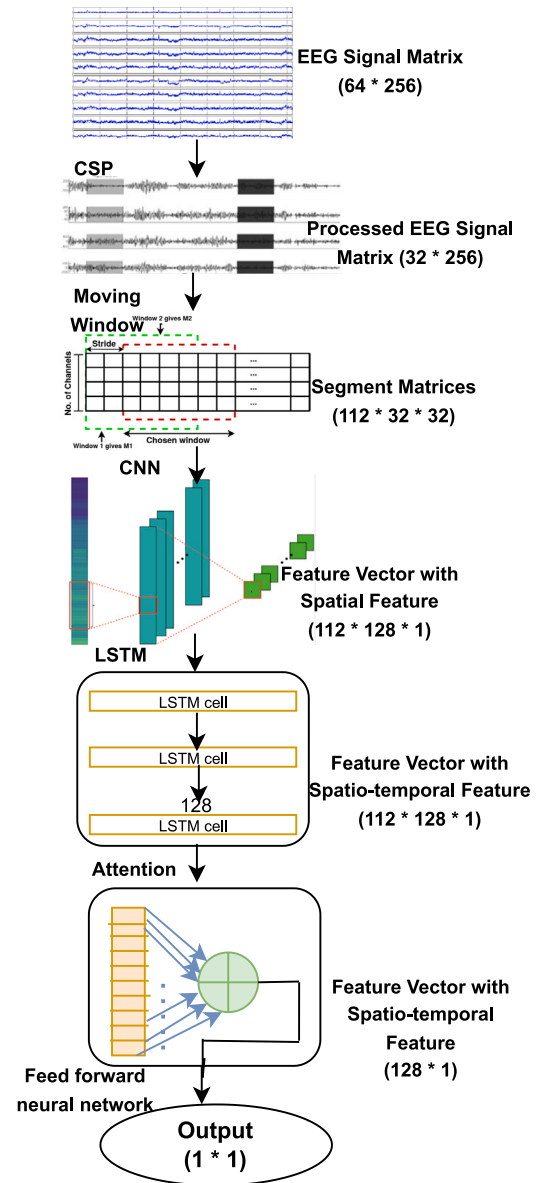


Fig. 7. Signal processing steps.

Furthermore, the Wilcoxon signed-rank test [25] is also used to prove the statistical significance of the proposed model. We use a significance level of $\alpha = 0.05$. Table 7 shows the test output for various state-of-the-art models compared to the proposed model. Many models shown in Table 5 used machine learning techniques for classification, as discussed in Section 1. These models have used statistical feature extraction techniques to extract handcrafted features. However, such approaches require domain-level knowledge and might not always generalize well. Many of these works have used very few samples to report the results, as discussed in Section 1. Moreover, the noise-prone, nonlinear, and non-stationary nature of EEG data makes statistical techniques and machine learning harder to apply, as most of these techniques presume the nature of the function that will map the input to the output and require the data to meet certain conditions. Deep learning models overcome these difficulties and make no such assumptions about the data. Deep learning models have been used for many nonlinear tasks and work well in several domains. However, the literature related to the classification of EEG signals for the detection of alcoholism lacks the application of deep learning models that can extract latent spatio-temporal characteristics and use these characteristics for classification.

Table 5
Comparison with existing methods.

Model	Classification technique	Features used	Model accuracy (%)
Kumar et al. [17]	SVM and fuzzy C-mean clustering	Absolute power, relative power and peak power frequency	88.00
Ehlers et al. [18]	Discriminant analysis classifier	Hurst exponent, correlation dimension, LLE and entropy	88.00
Shri et al. [5]	SVM and BPNN	ApEn ranked using ANOVA test scores	90.00
Fayyaz et al. [8]	LSTM	Features obtained from peak visualization method	90.00
Kannathal et al. [19]	Discriminant analysis classifier	Correlation dimensions	90.00
Acharya et al. [4]	SVM	LLE, SampEn, and some higher order spectra features	91.70
Faust et al. [6]	GMM, KNN, PNN, DT, FSC and NBC	Higher order spectra features	92.40
Farsi et al. [9]	ANN and LSTM	PCA with ANN and temporal features using LSTM.	93.00
Zhu et al. [7]	KNN and SVM	HVGE and SampEn	95.80
Proposed model	CNN-LSTM-ATTN	CSP and moving window	98.66

Table 6
Comparison with state of the art methods with their limitations.

Study	Features used	Classifier	Highest accuracy	Limitations
[20], 2020	Line-length (LL), log energy entropy (LEENT), and norm entropy (NENT)	LS-SVM, SVM, and k-NN	98.8	1. handcrafted features 2. partial dataset (120 signals from each class)
[21], 2021	NA	CNN	98	1. Partial dataset (30 samples from each class)
[22], 2015	Statistical features	RF	95.34	1. Partial dataset 2. Statistical feature
[23], 2017	Center correlated entropy	LS-SVM	97.02	1. Handcrafted features
[24], 2018	Log energy entropy features	LS-SVM	97.91	1. Handcrafted features
Proposed work	NA	CSP-CNN-Attention-LSTM	99.00	1. Complex architecture

Table 7
Wilcoxon signed-rank test results.

Models for comparison	Wilcoxon signed-rank test $\alpha = 0.05$; p-value
Proposed versus LS-SVM	0.047*
Proposed versus KNN	0.051*
Proposed versus RF	0.032*
Proposed versus CNN	0.036*
Proposed versus Regularized CNN	0.070

* Indicates that p value is less than or equal to 0.05.

Table 8
Performance of various variations of the proposed model(in %).

Model	Accuracy	F1-score	Recall	Precision	MCC	AUC
CSP-LSTM	84.83	84.95	85.66	84.26	69.67	92.83
CSP-LSTM-ATTN	93.66	93.62	93.00	94.25	87.34	98.12
CSP-CNN-LSTM	95.50	95.60	98.00	93.33	91.11	98.84
CSP-CNN-LSTM-ATTN	98.60	98.59	98.24	98.95	97.20	99.64

To our knowledge, the only works that used deep learning for this task are Fayyaz et al. [8], and Farsi et al. [9]. However, they applied simple deep learning models such as ANN and LSTM and focused more on the temporal features of the EEG signal and thus could achieve limited accuracy as reported in Table 5. The proposed architecture used a novel and hybrid deep learning architecture that extracts spatial and temporal features for classification and has shown high accuracy compared to other existing models proposed in the literature.

4.1. Ablation study

As discussed earlier, this architecture can take advantage of the fact that the EEG signals are spatio-temporal. This architecture extracts spatial and temporal features from the EEG signal and classifies them as alcoholic or control with high accuracy. To assess the contribution of each component in our proposed CSP-CNN-LSTM-ATTN model, we conducted an ablation study systematically removing different components and evaluating the model's performance. Table 8 presents the accuracy, F1-score, recall, precision, Matthews Correlation Coefficient (MCC), and AUC for different variations of the model.

CSP-LSTM: This baseline model applies CSP for feature extraction, followed by an LSTM for temporal modeling. It achieves an accuracy

of 84.83% with an F1-score of 84.95%, indicating that while LSTM captures temporal dependencies, the absence of CNN and attention limits spatial feature extraction and long-range dependencies.

CSP-LSTM-ATTN: Adding an attention mechanism significantly improves performance by increasing accuracy to 93.66%. Using the most pertinent temporal features, the attention module improves interpretability and raises MCC (87.34) and AUC (98.12). Spatial feature extraction is still limited, though, by the absence of CNN.

CSP-CNN-LSTM: Introducing a CNN before LSTM further improves performance, increasing accuracy to 95.50% and MCC to 91.11. CNNs effectively capture spatial dependencies, and their combination with LSTM enhances temporal feature extraction. However, the model cannot selectively focus on critical time steps without attention.

CSP-CNN-LSTM-ATTN: With an accuracy of 98.60% and an MCC of 97.20, the full model performs best across all metrics. The attention mechanism selectively highlights important features, the CNN records spatial features, and the LSTM models temporal dependencies, resulting in the highest F1-score (98.59%) and AUC (99.64). Table 8 shows that simpler models like LSTM or Attention along with CSP give good results, which are better than several existing techniques but perform worse than the initially proposed architecture. This table also shows that the model's performance will deteriorate if the attention layer is removed from the proposed architecture. Hence, from this study, it can be concluded that all the layers used in the proposed model are essential and non-redundant. Fig. 8 shows the Receiver Operating Characteristic (ROC) curve for all the variations of the propose discussed in Table 8.

4.2. Functional brain network analysis

We used network measures including degree centrality [26], clustering coefficient [27], and small-worldness [28] to examine functional brain networks of alcoholics and control participants. We consider two subjects (co2a0000372 (Alcoholic = True), co2c0000341 (Alcoholic = False)) for this analysis. The alcoholic participants had a lower degree of centrality (0.55 vs. 0.71), a smaller small-worldness (0.91 vs. 1.12), and a lower clustering coefficient (0.48 vs. 0.63) than the control subjects. According to these findings, there is less overall connectivity, worse local brain interactions, and less network efficiency in the alcoholic brain. The results align with prior research, suggesting that chronic alcohol consumption negatively impacts brain network organization. Table 9 summarizes the findings from the statistical

Table 9

Welch's t-test for functional brain network comparison metrics between control and alcoholic groups.

Metric	Control group (Mean $\pm$ SD)	Alcoholic group (Mean $\pm$ SD)	<i>t</i> -statistic	p-value
Degree centrality	0.71 $\pm$ 0.05	0.55 $\pm$ 0.06	3.42	0.002
Clustering coefficient	0.63 $\pm$ 0.04	0.48 $\pm$ 0.05	3.85	0.001
Small-worldness	1.12 $\pm$ 0.07	0.91 $\pm$ 0.06	3.67	0.0015

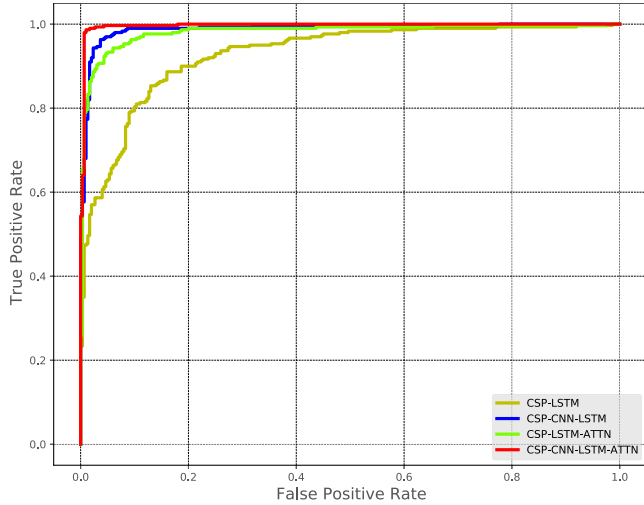


Fig. 8. ROC curves for variations of the proposed model.

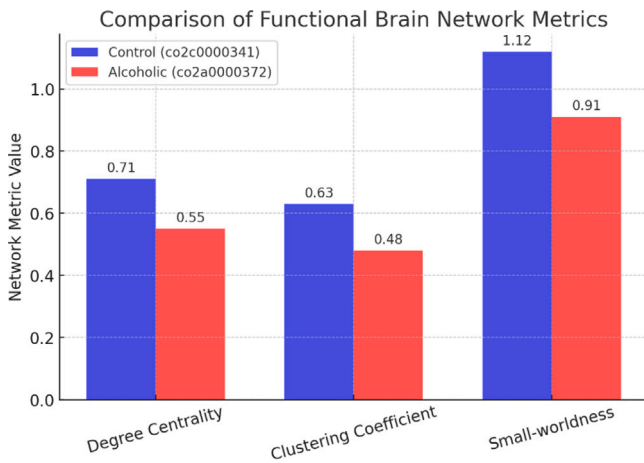


Fig. 9. Bar plot comparing the functional brain network metrics between the control subject (co2c0000341) and the alcoholic subject (co2a0000372).

significance assessment using Welch's t-test [2,29,30]. Lower p-values indicate significant differences between the groups, suggesting altered brain network organization in alcoholics. The differences between the groups are shown graphically in the appropriate bar plot 9.

Additionally, we have plotted four subgraphs for functional brain networks based on the Phase Lag Index (PLI) [31] and Pearson correlation to illustrate the primary differences in functional connection patterns between the alcoholic and control groups. Fig. 10 shows the comparison of functional brain networks in alcoholic and control groups using PLI and Pearson's correlation-based methods. In the PLI-based networks, the alcoholic group exhibits more localized connections, producing densely packed hubs, particularly in particular brain regions. This implies that maladaptive plasticity mechanisms may result from chronic alcohol consumption. Effective neural transmission is demonstrated by the control group's more balanced network of

Table 10

Power Spectral Density (PSD) values for different EEG frequency bands comparing the Control and Alcoholic groups. The trends indicate the relative increase or decrease in spectral power for alcoholics.

Band	Control group ($\mu V^2/Hz$)	Alcoholic group ($\mu V^2/Hz$)	Trend
Delta	4.6	7.0	↑ (Increased)
Theta	2.3	3.9	↑ (Increased)
Alpha	5.6	3.2	↓ (Reduced)
Beta	7.2	5.4	↓ (Reduced)
Gamma	3.1	1.9	↓ (Reduced)

local and global connections. The alcoholic group's decreased global connectivity in the Pearson correlation-based networks suggests less long-range functional communication between brain regions. This suggests a decline in cognitive ability, particularly in executive function and memory. The network of the control group stays evenly distributed and linked, maintaining optimal brain synchronization and communication. These modifications to the network architecture demonstrate how long-term alcohol consumption modifies brain chemistry, affecting cognitive and judgmental abilities.

4.3. Power spectral density (PSD) analysis

According to earlier studies on EEG-based alcoholism detection, the measured Power Spectral Density (PSD) values show significant variations between the control and alcoholic groups. Reduced alpha, beta, and gamma power in the alcoholic group suggest weakened neural synchronization, reduced attentional control, and worse cognitive function—all characteristics of chronic alcohol use. On the other hand, deficiencies in impulse control, neuronal slowness, and executive function are more likely to occur when theta power rises. These findings show how long-term alcohol use impairs brain function, particularly in regions related to memory, focus, and judgment. Using the UCI EEG Alcoholism dataset, we analyze PSD across frequency bands and observe trends consistent with prior research [32]. Representative PSD values for control and alcoholic groups, based on established findings, are summarized in Table 10.

A weakness of the current work is that some hyperparameter optimization of the neural network model is required to achieve the best model. However, this tuning of parameters and hyperparameters was limited to only CNN and LSTM models. Parameters are optimized using a grid search approach to explore the constrained parameter space and identify optimal configurations systematically. Also, due to the stochastic nature of initial weights and other model parameters, the results of the deep learning technique may vary in each training run, thus affecting the learning process. Combining CSP, CNN, attention mechanisms, and LSTMs results in a complex model with many parameters. This complexity increases computational burden and requires high-performance hardware (GPUs). Our implementation was done using Google Colab, which provides cloud-based computational resources. The experiments were carried out on the Google Colab free-tier GPU environment, which includes an NVIDIA Tesla T4 GPU (16 GB VRAM), 12 GB RAM, and a virtualized environment with Python 3.9 and TensorFlow/PyTorch frameworks. The model training time varied depending on the dataset size and hyperparameters. On average, training took approximately 122 s per epoch for 50 epochs.

5. Conclusion

The new architecture presented in this study combines statistical techniques like CSP with deep learning architectures such as CNN, LSTM, and Attention networks. This novel work uses this hybrid CSP-CNN-LSTM-ATTN network to classify EEG signals as alcoholic or control with high accuracy. This work shows how deep learning can extract spatiotemporal features from multichannel EEG signals and use these features for accurate classification tasks. There existed a gap

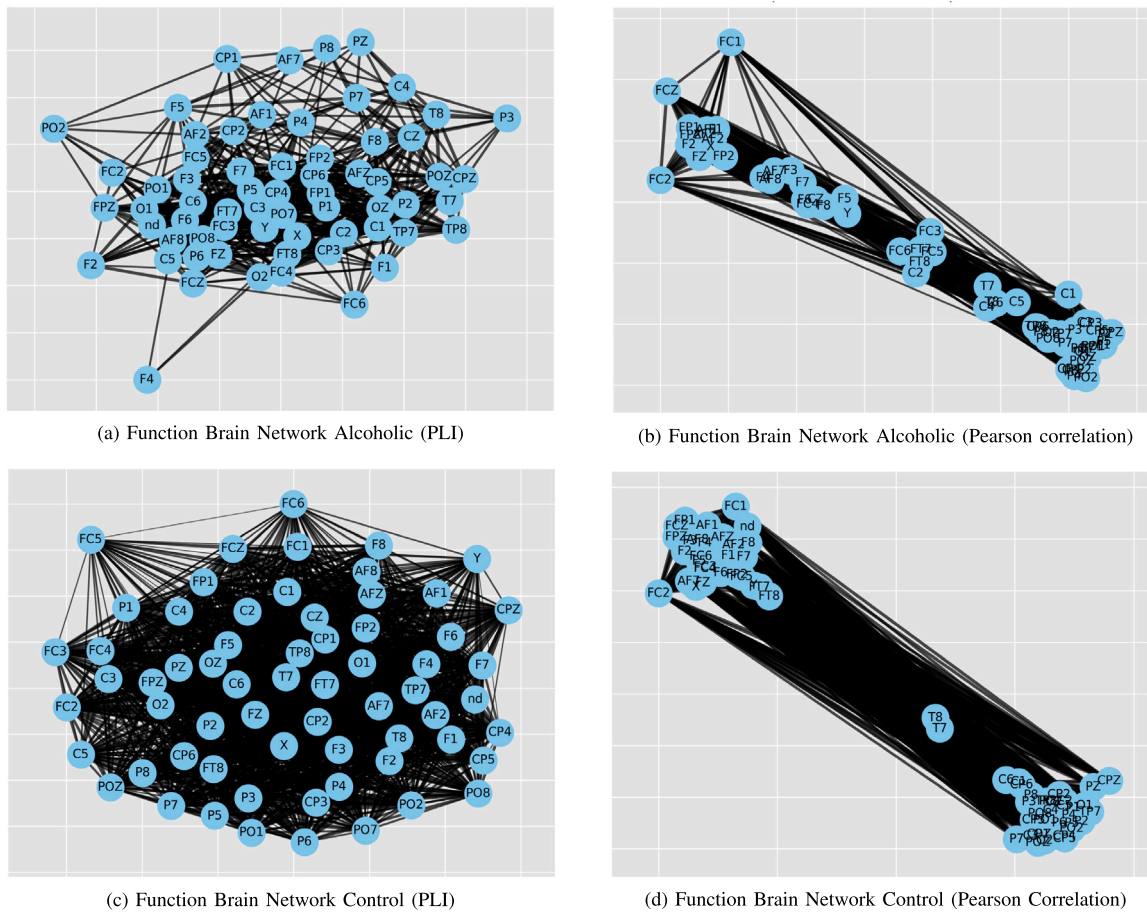


Fig. 10. Comparison of functional brain networks in alcoholic and control groups using PLI and Pearson correlation-based methods.

in the literature, as none of the existing works related to alcoholism detection using EEG signals used a hybrid deep learning architecture to extract spatio-temporal features from the EEG signal for classification purposes. Through this research, we aim to fill this gap in the literature. Moreover, the success of this model in analyzing EEG signals also makes this model a suitable candidate for various other EEG classification tasks. More broadly, this work can be extended to various tasks related to multichannel signal classification. Even though our method has produced encouraging results, more studies may examine using sophisticated deep learning architectures, including Transformers and Graph Neural Networks (GNNs), to increase the model's efficacy. GNNs may be helpful in situations where structured data relationships are essential. In contrast, transformers, renowned for their self-attention mechanisms, may improve the model's capacity to grasp long-range dependencies. Moreover, we will try to incorporate multi-modal data (EEG + MRI), using explainable AI techniques to enhance model interpretability, or validating results on larger, more diverse datasets.

CRedit authorship contribution statement

Neeraj: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Vatsal Singhal:** Methodology, Investigation, Formal analysis, Conceptualization. **Jimson Mathew:** Writing – review & editing, Supervision. **Ranjan Kumar Behera:** Writing – review & editing, Supervision.

Ethical statement

(1) This material is the authors' own original work, which has not been previously published elsewhere. (2) The paper is not currently being considered for publication elsewhere.

- (3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
- (4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
- (5) The results are appropriately placed in the context of prior and existing research.
- (6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.
- (7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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